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Multicast Requirements and Challenges in Super-Pod Applications

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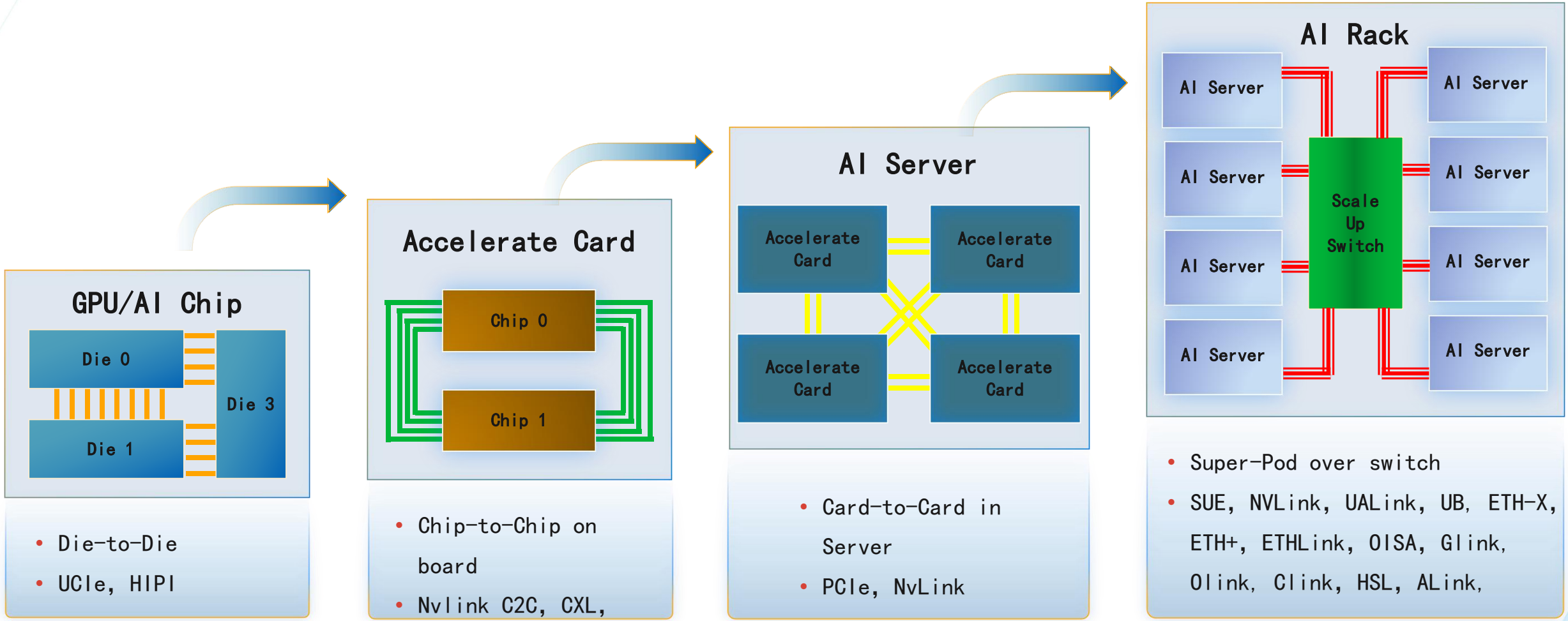
Shanghai UniVista Industrial Software Group Co.,Ltd.

Leading vendor of super-Pod interconnect IP

Multi-level interconnection drives super pod evolution



- Die-to-Die
- UCIe, HIPI

- Chip-to-Chip on board
- Nvlink C2C, CXL, Interlaken

- Card-to-Card in Server
- PCIe, NvLink

- Super-Pod over switch
- SUE, NVLink, UALink, UB, ETH-X, ETH+, ETHLink, OISA, Glink, Olink, Clink, HSL, ALink,

Multicast In-Network Computing Requirements in Super-Pod

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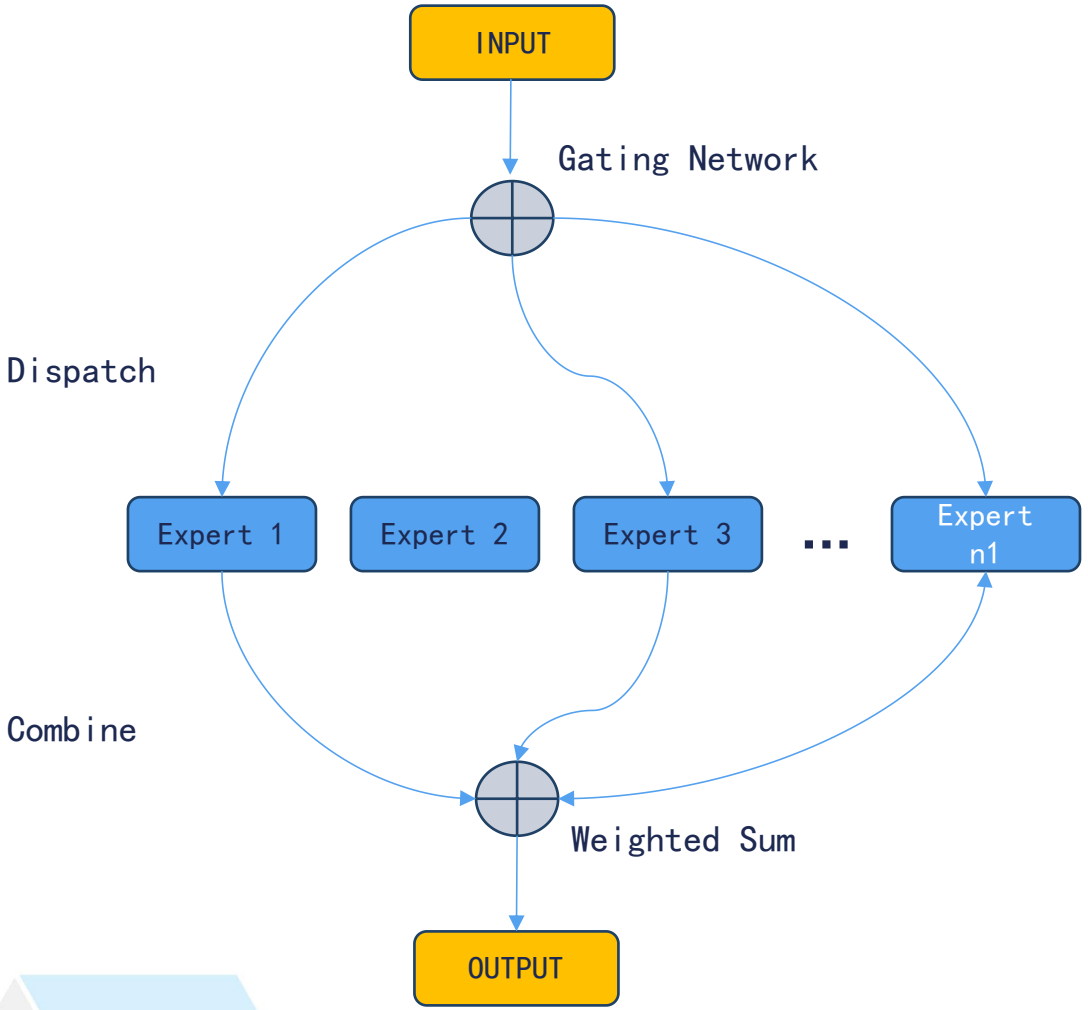
Optimize communication bottlenecks By Offloading computation to switches

- Less data movement
- Less bandwidth pressure
- Less power consumption
- Less latency

Improving effective computing utilization

- Less communication blocking and congestion
- Less idle state caused by computation waiting for data

MoE is a sparsely activated neural network.
It operates through the collaboration of multiple experts and a gating network, with each expert specializing in specific types of tasks.

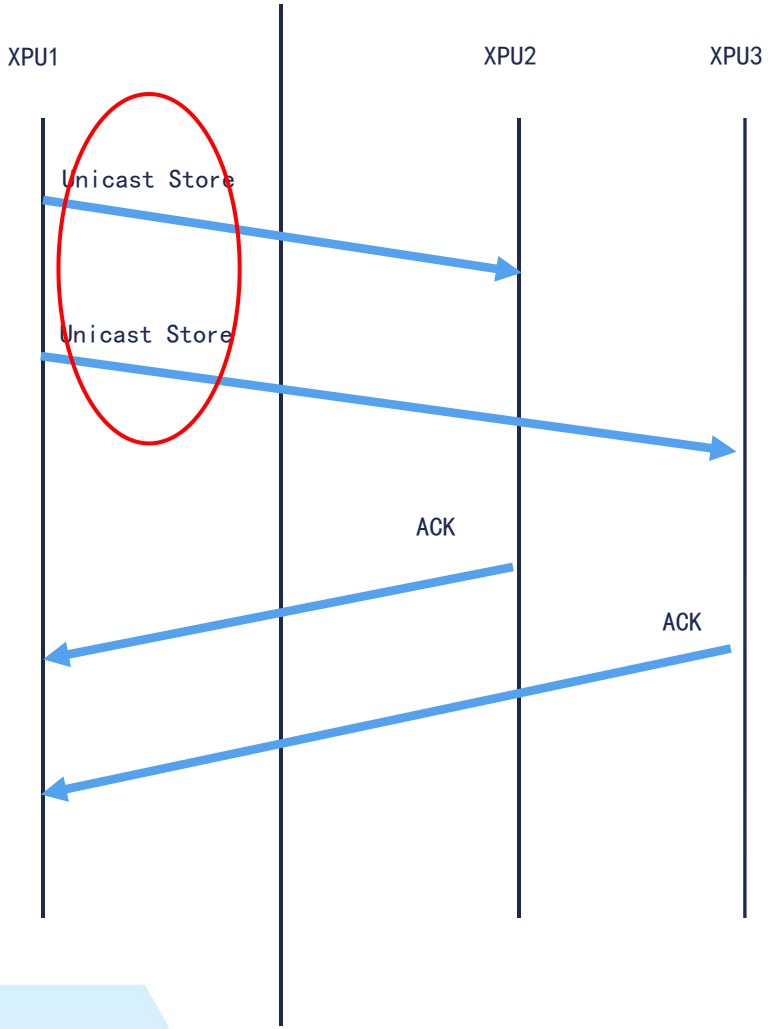


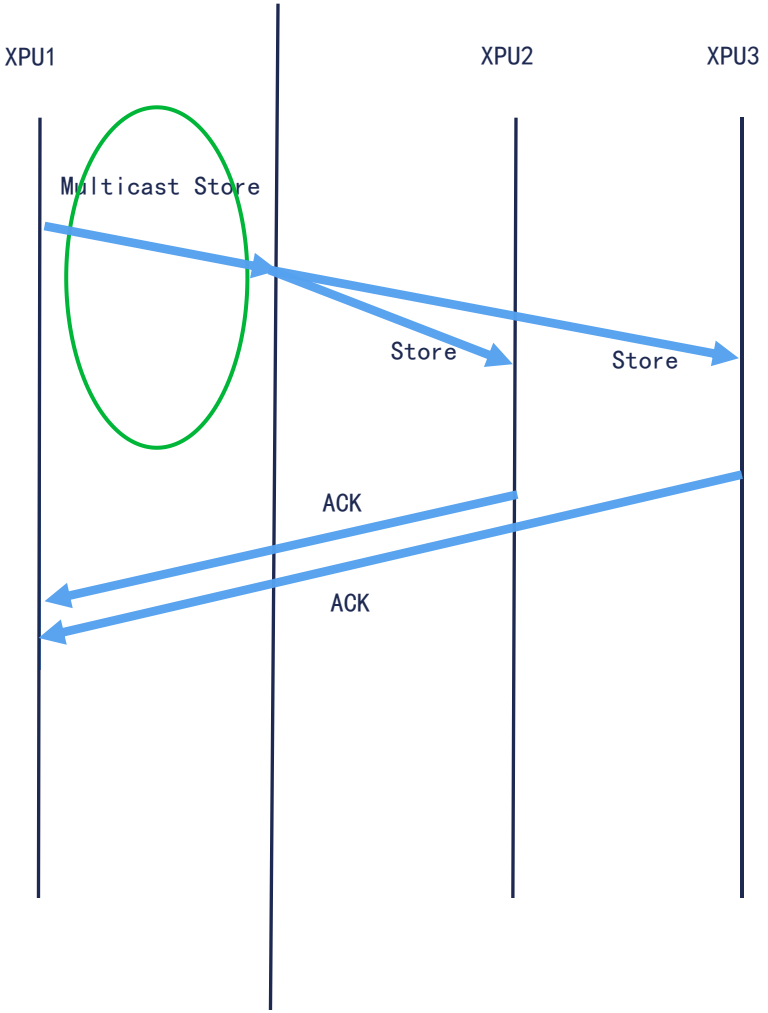
Typical flow of MOE

Dispatch Multicast information
Token content
Target expert GPU which host the TOP-K experts and corresponding weight
Target memory address for the combine operation

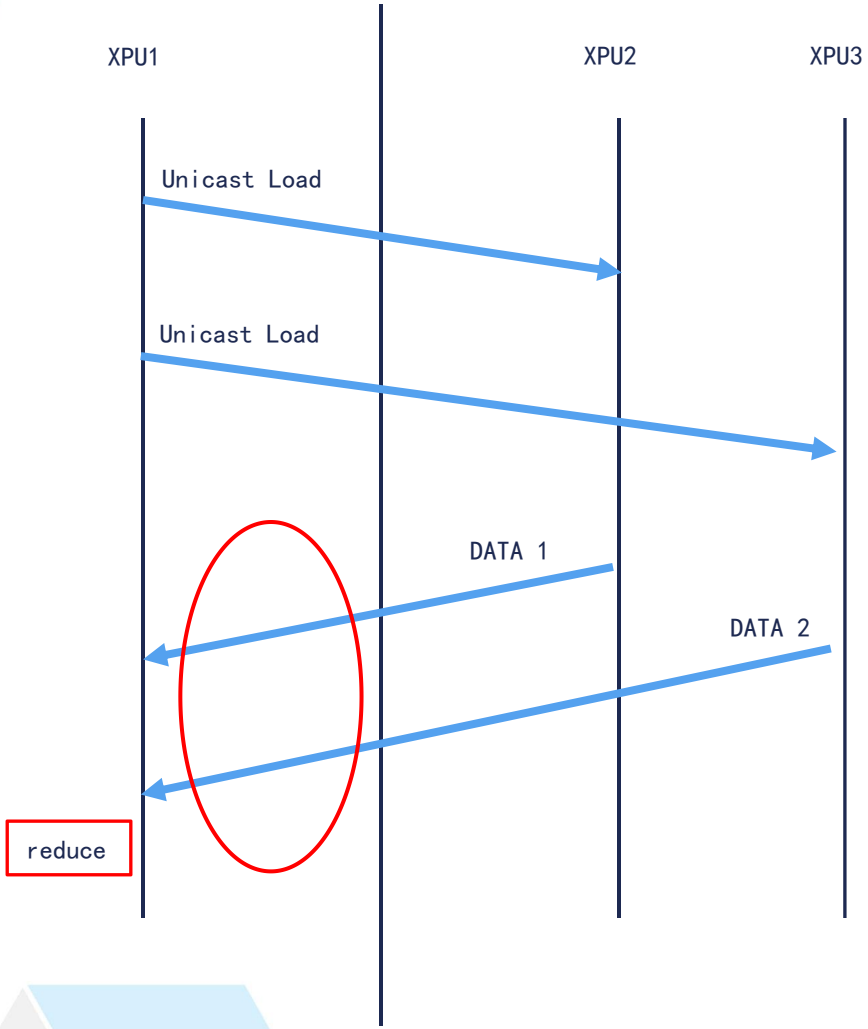
Combine
Aggregate the processing results of each expert with corresponding weight.

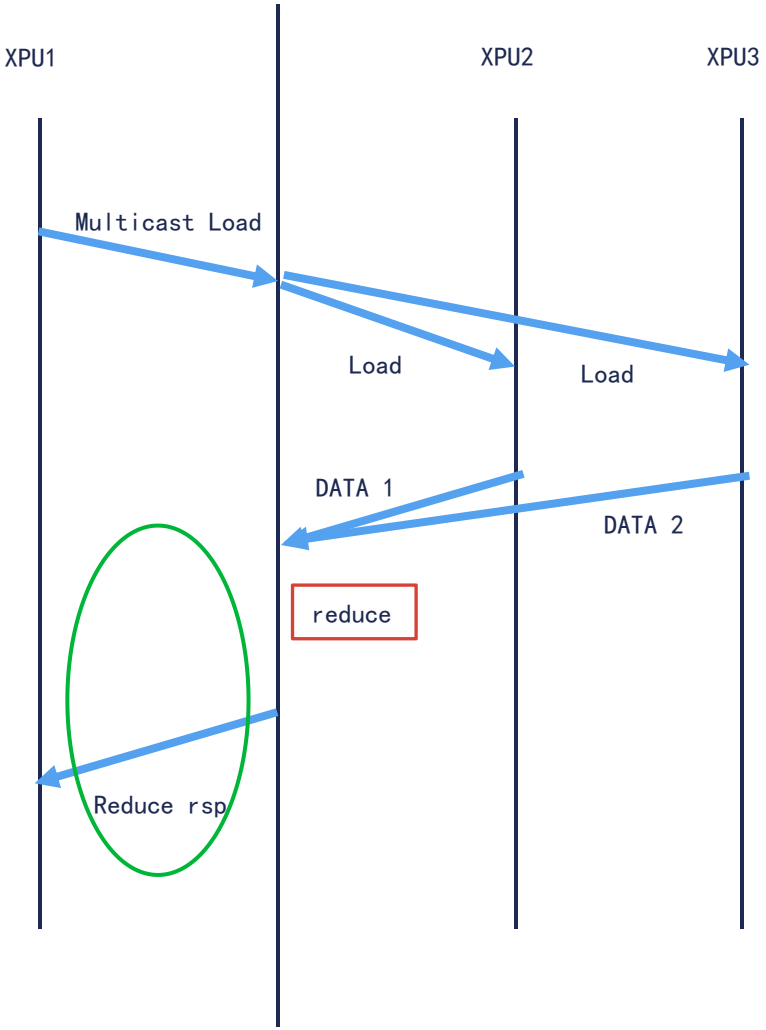
Dispatch: Unicast vs Multicast





Combine: Unicast vs Multicast





Multicast In-Network Computing Challenges in Super-Pod

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Ecosystem

Message semantics

based on Rocev2/IB, MOE not supported

Memory semantics, memory semantics are not unified

Full Transaction Layer should be implemented in switch, which protocol ?

Multicast behavior is no unified

Precision are not unified

FP32/FP16/FP8/BF16/TF32/INT8

Switch implementation

Dynamic multicast technology needed, Extremely large combinations

randomly activating 8 cards out of 256 in MOE, static multicast can’t support

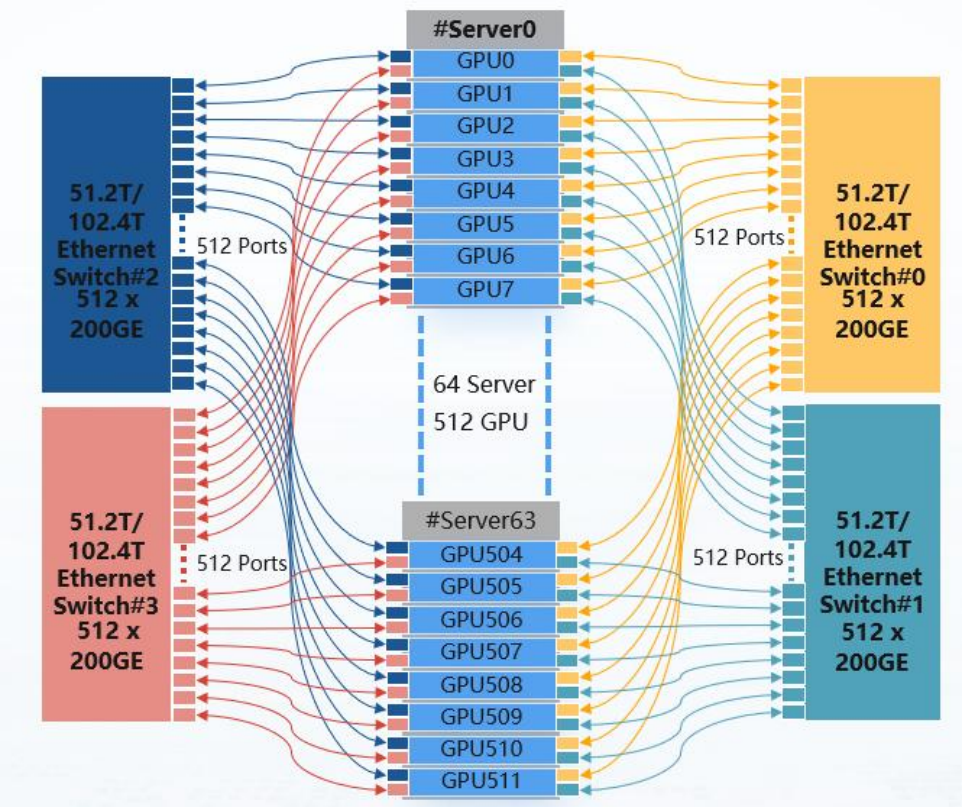
Larger bus bandwidth needed

extra traffic as Multicast

Memory resource gap

big gap between Inflight Number and actual usage

Multicast In-Network Computing Challenges in Super-Pod



Reliability

Optical transient interruption may cause LLR timeout.

Memory fault may occur.

E2E retry is difficulty in XPU, as the actual consumer of Multicast request are switches.

Limitation

Multi-layer topology are not supported

Load balance should be disabled, Network plane must be specified

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THANKS！

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